

A Hierarchical Digital Twin Network for Satellite Communication Networks

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ABSTRACT

Satellite communication networks promise to provide seamless global coverage in the 6G era. However, novel satellite networks face the challenges of high dynamics in terms of both communication and networking, which makes its design, emulation, deployment, and maintenance over-complicated. For these challenges, Digital Twin (DT) can establish a virtual replica of the physical system, providing a means of emulation, validation, prediction, and troubleshooting for the actual system whose benefits have been proven by many applications. In this work, we design a hierarchical digital twin network for satellite communication networks, which defines central-DT and edge-DT models that provide differentiated DT services such as communication DT and networking DT. Further, for the problem of model synchronization, we propose three approaches to improve the model synchronization efficiency, namely, dynamic DT migration, QoS-aware DT synchronization slice, and locator-identifier-isolation addressing mechanism. Simulation is conducted to validate the improvement of the proposed scheme, and provide proof of efficiency.

INTRODUCTION

The 6G communication is envisioned to provide ubiquitous broadband connectivity to users everywhere. However, limited by the return on investment, achieving global coverage with conventional terrestrial networks is not affordable[1]. Meanwhile, satellite communication networks have become a promising solution to provide global broadband connectivity in the 6G era[2]. Recently, many companies are vigorously developing satellite networks to provide data services in multiple aspects, including remote sensing, navigation, and post-disaster reconstruction[3]. Since deploying and operating a satellite network is expensive and complicated, advanced simulation and maintenance techniques have become imperative for system operators.

As an emerging concept, digital twin (DT) is envisioned as a promising solution for maintaining and operating physical systems (PS). DT continuously synchronizes with PS throughout its life cycle to construct an intelligent and evolving virtual replica [4]. With the support of artificial intelligence (AI), DT can build a high-fidelity model and perform precise predictions, which enables opera-

tors to make real-time network optimizations and operations. Hence, an elaborate DT system can complement PS in various aspects, such as design, deployment, and operation [5].

The advance of DT provides a promising solution for satellite communication networks. For example, in the deployment stage, the compatibility of the experimental satellites can be evaluated by emulating synchronously with the corresponding high-fidelity DT, and costly and irreversible mistakes can be avoided in advance [6]. In the operation and maintenance stage, operators can quickly diagnose network faults and make optimal decisions with DT's data collection and prediction capabilities.

Although the attractive vision DT promises, how to build DT for satellite communication networks remains an open issue. Among all challenges to constructing satellite networks' DT, the synchronization latency is an important issue. In satellite networks, online operations and fast reactions rely heavily on low-latency model synchronization. However, the long propagation path between DT models and satellite entities brings intense challenges for real-time model synchronization. Meanwhile, the topology dynamics may interrupt the connection between PS and DT. In dynamic satellite networks, DT must evolve continuously with PS as the physical entity moves and changes its environment [7]. Hence, the problems of large synchronization latency and network dynamics must be solved.

In this article, we present a novel hierarchical digital twin network for satellite communication networks (HDTN-SCN) to facilitate the deployment and evolution of satellite communication networks. We apply DT to build a virtual replicate of the satellite network, where operators can effectively develop and optimize the network with the help of software-defined network (SDN) and AI techniques. The main contributions of this work are summarized as follows.

We present a hierarchical architecture HDTN-SCN to provide differentiated data services for diverse applications. In the system, the edge-DTs are deployed in distributed ground stations to formulate physical entities for real-time services. Accordingly, the central-DTs are deployed in a centralized network control center (NCC) to facilitate global decision-making and optimization.

For the challenges when implementing the HDTN-SCN, we propose a regular and irregu-

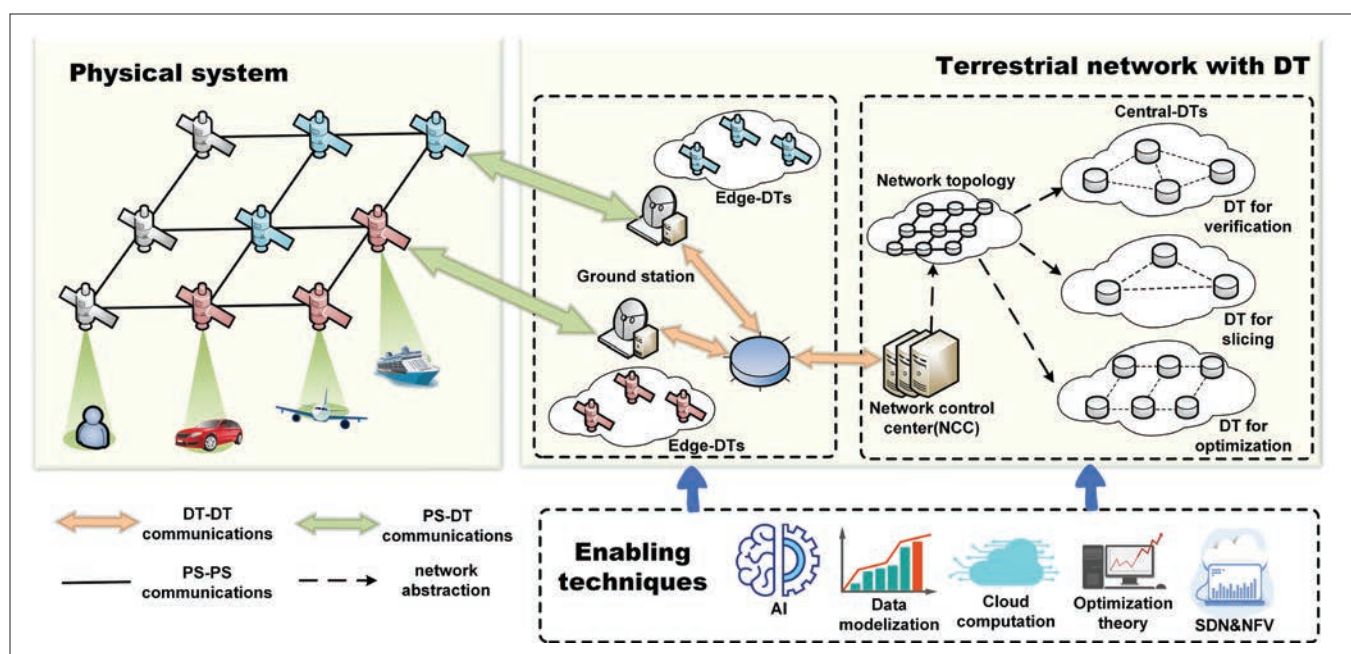


FIGURE 1. The system design of the hierarchical digital twin network for satellite communication network.

lar migration scheme for efficient DT migration, a QoS-aware slicing scheme for reliable PS-DT synchronization, and a locator-identifier-isolation mechanism for seamless DT addressing.

A case study on end-to-end path planning is presented to describe the system's workflow.

The performance of the proposed scheme is evaluated using computer simulations. The results show that HDTN-SCN outperforms the benchmarks of centralized architecture and the benchmark of hierarchical architecture without DT migration.

The rest of the article is organized as follows. In the following section, we introduce the recent advances of related works. Then we present the system design and its key components. A workflow is depicted in this section. Following that, we describe the networking challenges for the system. Next, we present the simulation results to show the performance of HDTN-SCN. Finally, we conclude the article.

RELATED WORKS

In this section, we present the recent advances in DT-enabled networks and networking issues in DT systems.

DT-ENABLED COMMUNICATION NETWORKS

Recently, some DT designs have been proposed to explore the combination of DT with future networks, including 5G[8], 6G[9], Internet of Vehicles (IoV) [10], air-space-ground integrated network (SAGIN)[11]. In [8], DT is envisioned as a highly interactive emulator in 5G automotive, 5G radio, and channel emulation. In addition, the authors proposed that DT can help 5G development, such as testing new core functions and validating the data rate with real-world traffic. [9] and [12] both proposed a DT-enabled 6G architecture to enable end-user virtualization with DT techniques. In [10], DT was integrated into vehicular networks to improve routing efficiency, maintenance, and traffic flow. In [11], DT is used to capture the dynamics of SATIN for training AI models.

Although DT has been introduced in several future network architectures, it is relatively new for satellite networks. Although satellite networks have the potential to provide broadband internet services in 6G [2], they still need to overcome the challenges of limited sources, harsh space environments, high dynamics, and time-varying latency. Significant research efforts are still needed in resource allocation, routing, topology reconfiguration, and mobility management [13]. DT can provide accurate models for network maintenance and operation and support developing and verifying new technologies, making it a potential tool for satellite network development.

NETWORKING ISSUES IN DT SYSTEMS

The networking support has been considered a key component to enable future DT implementations. Some contemporary works have explored networking issues. In [14], the authors discussed the importance of real-time bidirectional communication and state updates in a DT system. In [4], the authors discussed the key technologies of DT networks (DTN) and divided the communications of DTN into PS-PS, PS-DT, and DT-DT. In [15], the authors formulated the edge-DT association problem in dynamic wireless networks and decomposed it into DT placement and DT migration to solve it. In the existing works, the networking issues have been discussed in detail. However, the problems of large synchronization latency and network dynamics in satellite networks remain to be solved.

ARCHITECTURE OF HDTN-SCN

In this section, we describe the system design of HDTN-SCN. Then we present the critical components of HDTN-SCN, followed by a use case.

SYSTEM DESIGN

As shown in Fig. 1, the system is divided into a physical system and a terrestrial network with DT. The PS represents the satellite network composed of on-orbit satellites, wireless links, and

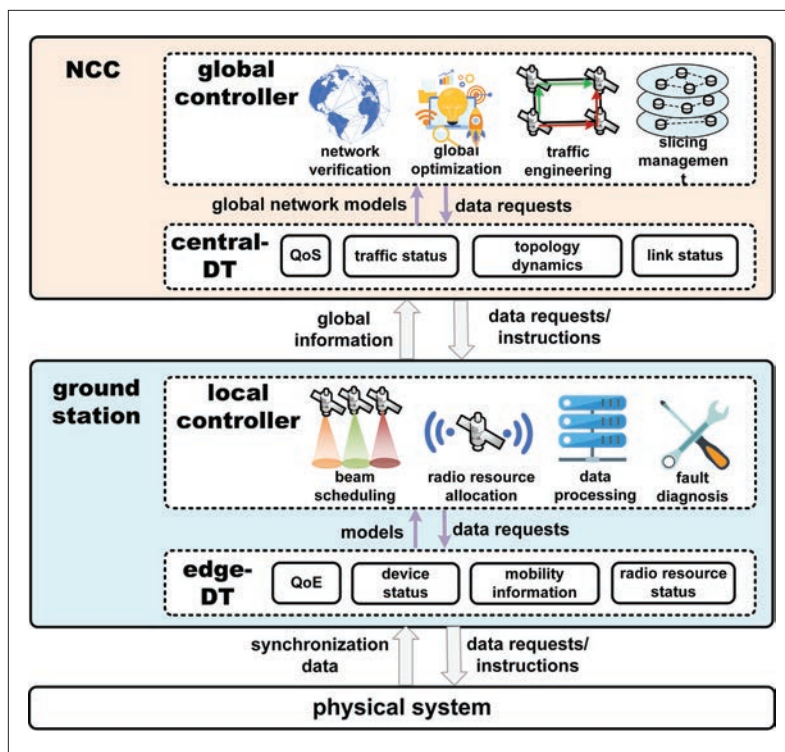


FIGURE 2. The function classifications of the DT system.

terminals. The satellites use wireless inter-satellite links (ISLs) to communicate with their adjacent satellites, while the terminals use ground-satellite links (GSLs) to access the satellite. In this case, the NCC controls the satellite network by communicating with geographically distributed ground stations. Each ground station can communicate with all physical entities through GSLs and ISLs. The terrestrial network connects ground stations and NCC, transmitting networking instructions and synchronization data for DTs and controllers.

Due to the limited satellite onboard resources, the DT system is integrated into the terrestrial network in a hierarchical architecture. In the system, each ground station maintains the edge-DTs of nearby physical entities. The local controller in the ground station can provide some local real-time network services based on the data models formulated by edge-DTs. The central-DTs are deployed in the NCC with a global SDN controller. The NCC collects global information from all edge-DTs and creates multiple central-DTs based on global information. The central-DTs can provide global network models for the global controller's networking applications. Since the satellite network is time-varying, the DTs can build AI models based on the current and historical data to capture the network dynamics and improve the modeling accuracy.

Note that DTs only focus on modeling with specific goals, while the controllers perform network operations based on DT models [12]. In the article, we mainly focus on the DT of satellites due to their specific characteristics, including high cost, dynamics, and high time-varying delay.

KEY COMPONENTS OF HDTN-SCN

In this subsection, we detail key components of HDTN-SCN, including edge-DTs, central-DTs, communications, and controllers. Furthermore,

we classify the functions of edge-DTs and central-DTs to clarify their differences.

Edge-DTs: In the system, edge-DTs are deployed on ground stations near its physical entities, as the colored satellites show in Fig. 1. Each ground station only maintains several edge-DTs. The edge-DTs request specific information from the physical entities following the prescribed data type, frequency, and precision. As shown in Fig. 2, the required information includes device status, mobility information, radio resource status, and Quality of Experience (QoE). Based on collected information, each edge-DT builds a model for the physical entity, which enables the local controller to perform real-time operations such as fault diagnosis, beam scheduling, radio resource allocation, and data processing. Deploying DTs in distributed ground stations rather than the centralized NCC can reduce the PS-DT synchronization latency, thereby improving the performance of most local operations. In addition, since simulating a high-fidelity DT requires massive data, processing data in ground stations can significantly reduce the transmission overhead of the terrestrial network.

Central-DTs: Although edge-DTs can provide low-latency data synchronization and local operations, a centralized compute-intensive node is indispensable for those global operations to aggregate the information collected from edge-DTs. Therefore, central-DTs are deployed in the NCC with a server cluster. As shown in Fig. 2, to construct the global network topology of the satellite network, NCC collects the required information from all edge-DTs, including traffic status, link status, QoS, and topology dynamics. NCC continuously maintains a global network topology. NCC abstracts the required information from the global topology and creates multiple isolated personalized central-DTs to meet diverse network management, and control requirements, including verification, global optimization, traffic engineering, and slicing management. In the system, one physical entity is only mapped to a unique edge-DT, whereas multiple central-DTs share edge-DT information. This method could significantly reduce the system overhead since the edge-DTs avoid redundant resource consumption from duplicating multiple DTs of the same physical entity for different central-DTs.

Controllers: Corresponding to the hierarchical structure of the DT system, there are local and global controllers in HDTN-SCN. The local controllers locate in ground stations and interface with the edge-DTs in the same station. As shown in Fig. 1, edge-DTs build models based on the modeling data collected by local controllers. Based on the edge-DTs and AI-based predictions, the local controllers can perform some local networking functions such as fault diagnosis and radio resource allocation. A global controller with sufficient resources is deployed in the NCC to provide various global networking functions. The global controller helps the central-DTs collect global information from the edge-DTs. With powerful computing ability, the controllers can build complicated AI models to predict the time-varying network conditions and optimize the networking schemes iteratively.

Communications in DT: As a fundamental component of the DT system, communications support the data exchange throughout the DT system. Com-

munications in DT can be decomposed into PS-PS, PS-DT, and DT-DT communications. The PS-PS communications represent the information exchange between all physical entities in the satellite network. PS-DT communications mean that a physical entity realizes data transmission with a DT in the terrestrial network through ISLs and GSLs and receives feedback from the DTs. The DT-DT communications represent the information exchange between different DTs. As shown in Fig. 1, DT-DT communications can be further divided into intra-DT and inter-DT communications. The inter-DT communications refer to the information interaction of edge-DTs in different ground stations and the synchronization between edge-DTs and central-DTs. Unlike communications that consume real transmission resources, intra-DT communications represent the virtual links inside the central-DTs, which use servers' computing capability to simulate the transmission behavior.

WORKFLOW: A CASE STUDY

This subsection provides a case study of HDTN-SCN to describe the system's workflow. Assuming that the NCC calculates an end-to-end routing scheme for a terminal with rigorous QoS requirements. The QoS requirement is a combination of multiple metrics, so the routing strategy is challenging to determine intuitively.

Data Synchronization: The edge-DTs synchronize with the satellite network following the prescribed data type, frequency, and precision. At the same time, the edge-DTs process the collected data and update their local model.

Topology Construction: The NCC collects traffic distribution information, topology dynamics, and other networking information from edge-DTs. Then the NCC maintains a global network topology reflecting the real-time network status.

Routing Strategies Selection: The NCC obtains the QoS requirements and selects K appropriate routing strategies for the following network path selection. These routing strategies could be heuristic or AI-based.

Topology Deduction: Based on the global topology, the NCC abstracts the required information and duplicates K central-DTs for each strategy. Based on the predictable movements of satellites and preset linking rules, each central-DT could easily deduce the following N topology snapshots.

Decision and Performance Evaluation: The NCC calculates the routing results for each routing strategy. The performance of each routing result is evaluated on N+1 network snapshots in parallel. The evaluating method could be an AI-based model whose inputs are routing results, traffic distribution, and topology snapshots. Then the strategy with optimal QoS performance is selected. The NCC generates control instructions and sends them to the network.

Monitoring and Improvement: Subsequently, the satellite network executes a new routing strategy. The edge-DTs continuously monitor the transmission performance and feed it back to the NCC to optimize the routing strategy.

Considering that it is difficult to determine the optimal strategy for solving a complex optimization, evaluating multiple strategies in parallel could help figure out an appropriate solution in an acceptable time. The flexible virtualization ability of DT enables parallel evaluation for multi-

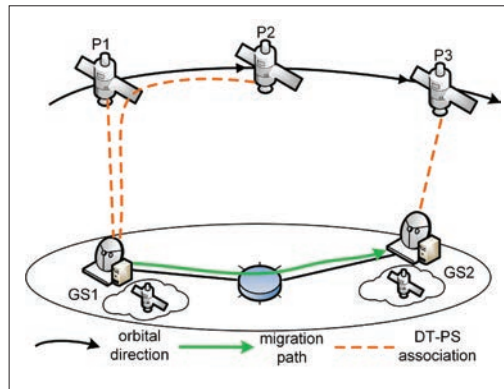


FIGURE 3. DT migration in HDTN-SCN.

ple schemes, which significantly reduces the processing latency. Furthermore, AI techniques could be further used to identify service features and improve the routing strategy.

IMPLEMENTATION ISSUES AND SOLUTIONS

There are some challenges associated with developing DTs in satellite networks. In the system, containers are adopted to realize the DTs of physical entities with high-fidelity requirements, such as satellites. For the large-scale terminals in the network, the DTs could be realized with more lightweight models. We present some networking challenges in HDTN-SCN and briefly discuss the solutions in this section.

DT MIGRATION

In the system, edge-DTs are placed in distributed ground stations to reduce the synchronization latency between DT and PS. Since the satellites operate at high speed in the orbits, the transmission latency between physical entities and ground stations is time-varying. To realize a low synchronization latency in the DT system, migrating edge-DTs of high dynamic entities between distributed ground stations is indispensable. As shown in Fig. 3, the edge-DT locates in GS1 when the satellite moves to P1. As the satellite moves from P1 to P2, the DT does not migrate since the transmission latency from P2 to GS1 does not exceed the preset latency threshold. When the satellite moves to P3, the latency between the satellite and GS1 exceeds the threshold, the NCC calculates the latency between the satellite and other ground stations and selects the nearest ground station GS2 as the migration destination. Next, the DT on GS1 migrates to GS2 through the terrestrial network.

Since frequent DT migration could incur considerable network overhead and increase the system's complexity, we propose a DT migration scheme taking full account of the features of the satellite network.

First, considering that the time-varying topology may incur frequent migrations, the synchronization latency threshold is introduced to reduce the migration frequency. Once the transmission latency of PS-DT exceeds the threshold, the edge-DT will migrate to a ground station with lower PS-DT latency. In addition, the load of ground stations should be considered in DT migration to avoid migrating to an overloaded station.

Second, considering the predictable movement of dynamic nodes, the DT migration is divid-

The global controller helps the central-DTs collect global information from the edge-DTs. With powerful computing ability, the controllers can build complicated AI models to predict the time-varying network conditions and optimize the networking schemes iteratively.

At each time stamp of migration, the edge-DT migrates to the destination and informs its physical entity that the association has updated.

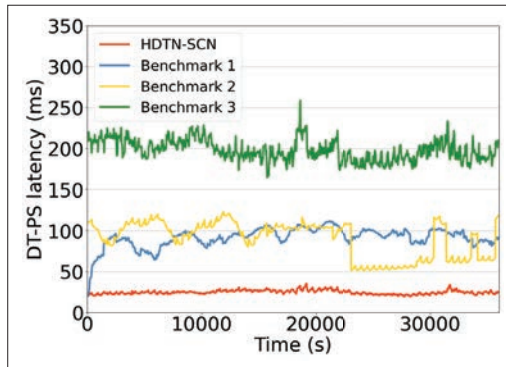


FIGURE 4. PS-DT latency vs. Time.

ed into regular migration and irregular migration. On the one side, the NCC plans the DT migration caused by the regular movement and presets the migration scheme on ground stations in advance. At each time stamp of migration, the edge-DT migrates to the destination and informs its physical entity that the association has updated. Additionally, regular migration can be improved by duplicating the DT. The physical entity can interact with the original DT until the duplicate is completely migrated to the destination and ready for modeling. This method makes the migration appear seamless to PS. On the other side, irregular migration is introduced to deal with network accidents, such as the considerable synchronization latency caused by satellite failures and the unavailable DTs caused by ground station failure. In this case, NCC detects the failure and plans a new migration scheme. Then the affected nodes are reconfigured. At the same time, NCC updates the global topology status and recalculates the regular migration plan for those affected DTs. Combining regular and irregular migration can effectively reduce the DT migration overhead.

RELIABLE PS-DT SYNCHRONIZATION

In a DT-enabled network, the difference between DT and PS may invalidate the DT and lead to unknown network accidents. Therefore, it is essential to realize reliable and low-latency PS-DT communications for accurate DT modelings.

However, achieving high-quality transmission of all DT data in a resource-limited satellite network is challenging. Differentiated communication can be adopted to guarantee reliable transmission for data services with diverse QoS requirements. For example, the data transmission for fault diagnosis must be low-latency, while the data for traffic engineering have lower QoS requirements. Network slicing has been adopted in 5G to provide differentiated transmission, which enables multiple logical networks over a shared physical infrastructure. Hence, we adopt a QoS-aware slicing scheme to meet the QoS requirements of DT synchronization. Three types of slices are considered, including low-latency slices, high-bandwidth slices, and best-effort slices. The DT data requiring low-latency reliable communication are transmitted through the low-latency slices, such as the monitoring data for default diagnosis and real-time instructions. The high-bandwidth slices are used for the DT data sensitive to reliability and bandwidth, such as the general DT modeling data. Although the DT system has rigorous

latency and reliability requirements, there still exist some delay-tolerant data that can be transmitted through best-effort slices. For example, in AI-based optimizations, the feedback information can be transmitted through best-effort slices since they are used to train the AI model to improve the accuracy of future decision-making.

Meanwhile, the controllers must monitor the achieved QoS levels of data services and adaptively adjust the slicing scheme. The slicing for the space segment must be adapted to the high dynamics, large propagation delay, and limited resources. In contrast, the slicing for the terrestrial segment can be mapped to the existing slices, such as eMBB and uRLLC.

ADDRESSING IN DT SYSTEM

In the proposed architecture, DT dynamic migration and the inherent high dynamics of satellite networks pose significant challenges to the network addressing. On the one hand, the centralized applications based on central-DTs must locate the corresponding DT according to the PS identity information. On the other hand, the location of PS should be quickly figured out based on DT's identifier.

We propose a locator-identifier-isolation mechanism for seamless DT addressing. The physical and digital objects are named by object identifiers, while the network addresses are used to locate the objects. The locator-identifier mapping table is introduced to support mobility, which maintains the mapping relationships between object identifiers and network addresses. There are multiple mapping tables in the system. First, each ground station must maintain a locator-identifier mapping for the DTs to locate corresponding physical entities. Meanwhile, considering the impact of DT migration, PS also needs to maintain a locator-identifier mapping of edge-DTs. These two mapping tables realize the data synchronization between DT and PS. Moreover, the NCC must maintain a PS-DT mapping table and a DT locator-identifier mapping table so that centralized applications can locate the corresponding DT through the identity of PS.

SIMULATION RESULTS AND DISCUSSIONS

In this section, we evaluate the proposed architecture and compare it with other potential designs. The performance evaluation consists of two parts: the PS-DT latency performance and the influence of the latency threshold.

SIMULATION SETTINGS

In the simulation, we focus on evaluating the DT performance of Low Earth Orbit (LEO) satellites since the LEO is the most dynamic entity in the satellite network. We conduct the simulations in two Iridium-like LEO constellations, namely con1 and con2. Con1 consists of 6 orbits with 11 satellites per orbit, while con2 consists of 12 orbits with 22 satellites per orbit. We select 23 geographically dispersed cities as the location of ground stations and choose Washington as the location of NCC. The NCC can communicate with ground stations through the terrestrial network, whose transmission latency is based on the real-time ping latency on <https://wondernetwork.com/pings>. In addition, the ground stations communicate with the satellites through GSLs and ISLs.

PS-DT LATENCY PERFORMANCE

In this subsection, we evaluate the PS-DT latency performance in Constellation A, which is defined as the propagation latency of the routing path between PS and DT. We propose three other potential system designs as benchmarks. Benchmark 1 is set to evaluate the hierarchical scheme with on migration. Benchmark 2 and benchmark 3 are set to evaluate the centralized scheme with different communication methods. The details are as follows.

HDTN-SCN: In this scheme, the PS-DT latency means the latency between the satellite and its corresponding edge-DT. Assuming that DTs will migrate seamlessly between ground stations. Therefore, each edge-DT is always located on a ground station close to the satellite.

Benchmark 1: The edge-DT of a satellite is initially deployed on its nearest ground station and will not migrate anymore. When the satellite moves away from its edge-DT, it must communicate with edge-DT through more satellite links.

Benchmark 2: Assuming that all DTs are deployed in the NCC. The satellite directly communicates with the NCC through the shortest path composed of satellite links.

Benchmark 3: Assuming that all DTs are deployed in the NCC. Unlike Benchmark 2, the satellite sends the synchronization data to the nearest ground station to save limited satellite network resources. Then, the ground station transmits it to the NCC through terrestrial networks.)

In Fig. 4, we compare the PS-DT latency performance of the above four schemes in 36,000 seconds, which is almost six orbital periods. As shown in the figure, the HDTN-SCN outperforms the benchmarks with an average latency of about 24 ms because the edge-DTs are always located at the ground station near the satellites, significantly reducing the propagation latency. In contrast, for Benchmark 1, the average latency increases to 92 ms without DT migration. The average latency of Benchmark 2 is about 90 ms in a centralized architecture. This is because the satellite can communicate with the DT directly through the optimal path of satellite links. However, the massive DT synchronization data may consume considerable resources and aggravate the network congestion around the NCC. A more economical way is to offload the data to the nearest ground station and transmit them to the NCC through the terrestrial network. As Benchmark 3 shows, its average latency is up to 200 ms due to the larger latency of the terrestrial network.

HDTN-SCN notably outperforms the benchmarks in PS-DT latency, enabling low-latency local operations with edge-DTs. Nonetheless, the DT system still requires a centralized node to aggregate global information. Therefore, we evaluate the latency performance in the centralized architecture with minimum latency and minimum ISLs, as Benchmark 2 and Benchmark 3 show. It is necessary to provide differentiated transmission services for diverse global information. For example, it is better to transmit data with rigorous real-time requirements through the optimal path composed of ISLs, while the delay-tolerant data are transmitted through the paths with minimized ISLs.

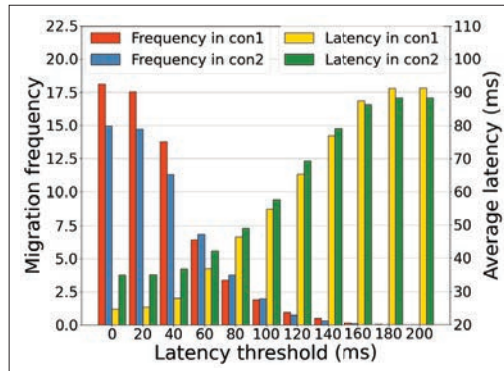


FIGURE 5. Migration frequency and average latency vs. latency threshold.

INFLUENCE OF LATENCY THRESHOLD

Although frequent DT migration ensures low PS-DT latency, it also incurs extra overhead. Therefore, we evaluate the latency threshold's impact on the migration frequency and average PS-DT latency. The migration frequency represents the average migration times for an edge-DT in almost 6000 seconds. We perform the simulations in two constellations.

As is shown in Fig. 5, as the threshold rises from 0 ms to 200 ms, the migration frequency declines gradually while the latency increases. When the threshold is 0 ms, the DT system intends to realize the minimum PS-DT latency with frequent migration. In contrast, when the threshold is 200 ms, the DT migration frequency equals zero since the propagation latency of the satellite network will not exceed 200 ms.

For constellation 1, when the threshold is set to 120 ms, the results mean that migrating once per period can realize a 27% reduction in PS-DT latency. When the threshold is set to 80 ms, the results mean that migrating three times per period can realize a 50% reduction in PS-DT latency. Therefore, we can set an appropriate latency threshold to realize a considerable latency reduction with acceptable DT migration frequency. In addition, similar conclusions can be drawn from the simulation results of Constellation 2. The results show that a considerable improvement can be realized even in a low migration frequency, which can guide designing a high-performance DT migration scheme based on complex factors in a heuristic or AI-based method in the future.

CONCLUSIONS AND FUTURE WORKS

In this article, we have proposed a hierarchical digital twin network for satellite communications to facilitate the deployment and evolution of satellite networks. We have designed the edge-DT and central-DT to provide differentiated data services for network operations. We have also presented some networking-related implementation issues in HDTN-SCN and briefly discussed the solutions. The evaluation results have shown that the proposed schemes outperform general schemes.

There are still several works to be done in the future:

- The DT migration scheme can be further developed. We have proved the magnificent latency performance realized by DT migration. However, the data interaction and modeling related to DT migration need further development to realize cost-effective

In contrast, for Benchmark 1, the average latency increases to 92 ms without DT migration. The average latency of Benchmark 2 is about 90 ms in a centralized architecture.

Although frequent DT migration ensures low PS-DT latency, it also incurs extra overhead. Therefore, we evaluate the latency threshold's impact on the migration frequency and average PS-DT latency.

DT migration. We have proposed a latency threshold and a DT migration scheme to reduce migration frequency. Next, we plan to elaborate on seamless migration and propose an AI-based migrating algorithm to determine the optimal migrating destination.

- Advanced networking techniques are required for real-time synchronization. We have proposed a QoS-aware slicing mechanism to provide differentiated transmission for diverse synchronization data. However, it is challenging to keep tight synchronization for high-priority data. Note that time-sensitive networking (TSN) is envisioned as a promising solution for realizing bounded end-to-end latency in terrestrial networks. Hence, we plan to integrate TSN with the proposed architecture. 3) the combination of AI and DT-enabled satellite networks can be further investigated. AI pervades the entire architecture and plays a critical role in DT systems in modeling, prediction, and optimization. Hence, we plan to apply AI techniques in HDTN-SCN to optimize resource allocation and routing in the next step.

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